The Toda two-body edge phase at fixed dressed momenta: directional one-loop coefficient, exchange decomposition, and the surviving physical-normalization question

Research note for Gustav Delius

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Abstract

(c) Conditional on the corrected half-line asymptotics, the inherited *two-term directional* Toda phase has the following leading small-amplitude, fixed-dressed-momentum correction. For right-moving overtaking solitons, $r = \beta_2/\beta_1 \in (0,1)$,

$$R_{12}^{(1)} = -\frac{4}{\pi}[\chi_2(r) - r],$$
$$S^{(1)} = \frac{2}{\pi}(r + r^{-1}) - \pi,$$
$$\vartheta_N^{(1)} = \pi - \frac{4}{\pi}\chi_2(r) + \frac{2}{\pi}(r - r^{-1}),$$

where $S = (R_{12} + R_{21})/2$, $\vartheta_N = (R_{12} - R_{21})/2$, and $\chi_2(r) = [\mathrm{Li}_2(r) - \mathrm{Li}_2(-r)]/2$. The reverse ordered branch is supplied by exchange, not by substituting $r > 1$ into principal real logarithms. The corresponding leading opposite-edge directional coefficients vanish. New finite-sea solves, high-precision inversions and unconstrained intercept fits test these statements; the numerical budgets are given below.

(c) The same momentum logarithm fixes the symmetric term and the one-body invariant $C - s_1 = -1/(4\pi)$. We also find a two-term/four-term mismatch in the Krein equation printed in the frame audit. Its correction does not restore an arbitrary antisymmetric Toda freedom. **(d)** These results do *not* identify ϑ_N with the Gamma-relative continuum residual, nor establish a uniform fixed-nonzero-momentum semiclassical limit. The exact endpoint-bilinear representative has additional one-body reference terms, including half-integer powers in the edge expansion. They are explicitly retained. Thus the literal fully normalized physical scalar phase requested in the task is not established by this package.

1 Scope, conventions and the observable actually computed

Labels retain the repository meanings: (a) proved in literature, (b) literature heuristic, (c) project derivation or numerical evidence, (d) unestablished interpretation. No external paper, abstract, arXiv entry or DOI is cited or claimed newly read. Internal sources are listed at the end. All new scripts and caches have the required `twobody_toda` prefix. No earlier script is run or imported, and no git state is touched.

(c) Retain the audited units $a = 1$, $b = e^{-1/2}$,

$$h = \hbar_T = 3b\hbar_B, \qquad \kappa = \sqrt{3}\beta = \sqrt{hy/b}, \qquad y = \beta^2/\hbar_B, \qquad P_B = \overline{P}/(6\sqrt{3}b). \tag{1}$$

The inherited continuum is the bad equation $u_{tt} = u_{xx} + u_{xxxx} + 3(u^2)_{xx}$, $u = 2(\log\tau)_{xx}$, $k = \sinh\beta$, $v = \epsilon\cosh\beta$. The dictionary is a small-amplitude dictionary; no finite-amplitude Toda–Boussinesq equivalence is inferred. In physical Boussinesq momenta the leading ratio is $r = (|P_{B,2}|/|P_{B,1}|)^{1/3}$, since $|P_B| = 2\beta^3/3 + O(\beta^5)$ in this corner. The exact classical Toda quantities used in the numerical

subtractions are

$$P_0(\kappa,\epsilon) = 4\epsilon b(\kappa\cosh\kappa - \sinh\kappa), \qquad p_0 = 2\epsilon b\cosh\kappa, \tag{2}$$

$$M_0 = -2\kappa, \qquad \Pi_0 = 2\epsilon b\sinh\kappa, \qquad v_0 = \epsilon b\sinh\kappa/\kappa. \tag{3}$$

At each finite h we solve $\overline{P}(p_j) = P_0(\kappa_j,\epsilon_j)$; we never replace the rapidity coordinate by a dressed momentum.

(c) Set $e_j = \epsilon_j q$ and define the directional phase and its four-term counterpart by

$$N_{12} = \Phi(p_1,p_2) - \Phi(p_1,e_2), \tag{4}$$

$$U_{12} = \Phi(e_1,p_2) - \Phi(e_1,e_2), \qquad \widehat{\Phi}_{12} = N_{12} - U_{12}. \tag{5}$$

Our residual is precisely the audited Phase 2f/2h two-term quantity, with the *full classical lattice* subtraction:

$$R_{12,h} = \frac{N_{12} - \delta_T(1,2) - C_{12}}{h}, \qquad C_{12} = M_{0,1}(P_{0,2} - \Pi_{0,2}). \tag{6}$$

Here $\delta_T = (p_{0,1}-p_{0,2})\log A_T + M_{0,1}M_{0,2}(v_{0,1}-v_{0,2})$, with $A_T = \sinh^2[(\kappa_1-\kappa_2)/2]/\sinh^2[(\kappa_1+\kappa_2)/2]$ for equal directions and the same ratio of $\cosh^2$ for opposite directions. The subtraction is evaluated with 60-digit arithmetic before comparison to the float64 sea solves. The dimensionless action phase is unchanged by the audited action rescaling: $\delta_T/h = \delta_B/\hbar_B$ in the scaling limit.

(d) A directional thermodynamic Bethe phase need not be antisymmetric. Consequently the notation ϑ_N is deliberate: it denotes the exchange-odd part of (6), not an already normalized scalar S-matrix, and not the residual relative to a Gamma candidate in `smx_candidates.tex`. This distinction matters especially for head-on scattering and absolute phase constants.

Checks: `twobody_toda_scan.py`.

Checks: `twobody_toda_identities.py`.

2 Exact finite-sea identities and the reference term

(c) With $t = 2\pi h$,

$$\Theta(d) = 2h\,\mathrm{Im}\log\Gamma(1+id/h) + 2d\log h, \qquad K(d) = \Theta'(d), \tag{7}$$

$$(\mathcal{A}f)(x) = tf(x) - \int_{-q}^{q} K(x-z)f(z)\mathrm{d}z, \qquad \mathcal{A}\rho = 1, \quad \mathcal{A}\eta = x, \quad \int\rho = 1, \tag{8}$$

$$\mathcal{A}F_z(x) = \Theta(z-x), \qquad \Phi(p,z) = \Theta(p-z) - \int_{-q}^{q} K(p-x)F_z(x)\mathrm{d}x. \tag{9}$$

The gamma logarithm is continuously unwrapped on the real difference axis. Let $D = P'$, $V = E'$, $D_q = t\rho(q)$, $V_q = t\eta(q)$. The audited endpoint identities and Jacobian are

$$D_q[F_p(q) - F_p(-q)] = 1 - D(p), \tag{10}$$

$$V_q[F_p(q) + F_p(-q)] = P(p) - V(p), \tag{11}$$

$$\frac{\mathrm{d}u}{\mathrm{d}y} = \frac{2\sinh\kappa}{D(q+hu)}, \qquad \overline{P}(q+hu) = P_0(\sqrt{hy/b}). \tag{12}$$

The high-precision checker also uses an independently counted spacing $a \neq 1$, restoring a in the Jacobian denominator, and differentiates the inverted momentum numerically. Its largest $N = 48$ Fredholm identity residual is $2.57\,10^{-30}$; the centered-difference Jacobian error is $6.65\,10^{-22}$.

(c) Evaluating the defining equation for F_z at an endpoint gives the particularly useful identity

$$\boxed{\Phi(e,z) = -tF_z(e), \qquad U_{12} = -t[F_{p_2}(e_1) - F_{e_2}(e_1)].} \tag{13}$$

This fixes the one-body reference term without a fit or a phase convention. Writing $f_j^{\pm} = F_{p_j}(\pm q) - F_{e_j}(\pm q)$, the exact reciprocity law is

$$\widehat{\Phi}_{12} + \widehat{\Phi}_{21} = t(f_1^+ f_2^+ - f_1^- f_2^-) = \mathfrak{M}_1\mathfrak{Q}_2 + \mathfrak{M}_2\mathfrak{Q}_1, \tag{14}$$

where $\mathfrak{M}_j = D_q(f_j^+ - f_j^-)$ and $\mathfrak{Q}_j = t(f_j^+ + f_j^-)/(2D_q)$ at $a = 1$. The independent checker solves the Fredholm equations and compares actual forward and reverse phases to this product, in both channels.
Checks: `twobody_toda_identities.py`.
Checks: `twobody_toda_scan.py`.

3 Correct half-line solution and a new reference correction

(c) Use only the corrected Wiener–Hopf normalization. Define

$$g(s) = s^{-1/2}\Gamma(1 + s/(2\pi)) \exp\{-s \log[s/(2\pi e)]/(2\pi)\}, \qquad \mathcal{L}G = \sqrt{\pi}\, g(s)/s^2. \tag{15}$$

Put $G(u) = \frac{4}{3}y(u)^{3/2}$, $B = G'$, $f = G''$, $c = \sqrt{\pi}$. Then $B(0^+) = c$, $\mathcal{L}B = \sqrt{\pi}g/s$ and $\mathcal{L}f = \sqrt{\pi}(g - 1)$ for the ordinary derivative on $u > 0$; the removed constant is the contact delta. The dressed Jacobian reads $B(u(y)) = 2\sqrt{y}/u'(y)$.

(c) For $u > v > 0$ the corrected, audited forward convolution is

$$H(u, v) := \lim_{h\to 0} N_{12}/h = 2\int_0^v f(u - w)B(v - w)\mathrm{d}w. \tag{16}$$

Equation (13) and the endpoint asymptotics $f_j^+ \to 1 - B(u_j)/c$, $f_j^- \to 0$ give

$$U(v) := \lim U_{12}/h = 2c[B(v) - c], \tag{17}$$

$$\widehat{H}(u, v) = H(u, v) - U(v), \tag{18}$$

$$\widehat{H}(u, v) + \widehat{H}(v, u) = 2[B(u) - c][B(v) - c], \tag{19}$$

$$\boxed{H(u, v) + H(v, u) = 2B(u)B(v) - 2\pi.} \tag{20}$$

The -2π must not be dropped. In particular $H(u, u) = B(u)^2 - \pi$; the four-term phase vanishes on either edge.

(c) There is a further, independently checkable correction to the frame audit. The directional convolution and the four-term function satisfy

$$(\partial_u + \partial_v)H = 2f(u)B(v), \tag{21}$$

$$(\partial_u + \partial_v)\widehat{H} = 2f(u)B(v) - 2cf(v). \tag{22}$$

The frame audit instead prints $2f(u)[B(v) - c]$ for the latter source. Its difference from the correct source is $2c[f(u) - f(v)]$, generally nonzero. This is a reference mismatch, not a new homogeneous freedom. Given the correct source and axis data, the characteristic equation still fixes the phase. The exact endpoint identity, finite-sea tests, symbolic bookkeeping and differentiated convolution are used, rather than the incorrect printed four-term PDE. No conclusion that arbitrary two-body additions are allowed is drawn from this correction. At $(u, v) = (4, 1)$, independent fourth-order differentiation of the convolution gives -0.347104254662920476, agreeing with the corrected source to $2.07\,10^{-17}$; the printed source is $+0.837748060692060127$ and differs by 1.18485231535498060.
Checks: `twobody_toda_identities.py`.
Checks: `twobody_toda_edge.py`.
Checks: `twobody_toda_analysis.py`.

4 Extracting the one-loop coefficient at fixed dressed momentum

(c) Denote $y_1 = Y$, $y_2 = r^2 Y$, $0 < r < 1$. First take $h \to 0$ at fixed (Y, r), then $Y \to \infty$. Each finite-h momentum is exactly inverted; the classical edge phase is

$$H_{\rm cl} = 4rY + 2Y(1-r^2)\log\frac{1-r}{1+r}. \tag{23}$$

At finite Y, $H - H_{\rm cl}$ contains higher inverse-Y corrections; it is not by itself a pure $O(\hbar^0)$ coefficient.

(c) The corrected momentum expansion has

$$u(y) = y - A\log y + c_{\rm edge} + O(y^{-1}), \qquad A = \frac{1}{2\pi}, \qquad B(u(y)) = 2\sqrt{y}\,[1 + A/y + O(y^{-2})]. \tag{24}$$

The constant $c_{\rm edge}$ cancels from the leading two-body correction. To see the cancellation, substitute $w = v - u(z)$ in (16):

$$H = 4\int_0^{r^2 Y} \sqrt{z}\, f\big(u(Y) - u(r^2 Y) + u(z)\big)\mathrm{d}z. \tag{25}$$

With $S = Y - r^2 Y + z$, expansion of the integrand at fixed positive z/Y gives $f = S^{-1/2} - \frac{A}{2} S^{-3/2}\log[Sr^2/z] + \cdots$. Thus the leading residual reduces to

$$R_{12}^{(1)} = -2A\int_0^{r^2} \frac{\sqrt{t}}{(1-r^2+t)^{3/2}}\log\frac{(1-r^2+t)r^2}{t}\mathrm{d}t \tag{26}$$

$$= 8A\int_0^{r} \log(s/r)\frac{s^2}{1-s^2}\mathrm{d}s = -8A[\chi_2(r) - r]. \tag{27}$$

The change of variable is $s = \sqrt{t/(1-r^2+t)}$. The integrals and their equality to the dilogarithms are checked at 70 digits. **(d)** The expansion at $z = O(1)$ requires a matched endpoint estimate to turn this into a uniform theorem for the finite-sea equation. Direct inversion and unconstrained fits test the result without assuming its intercept.

(c) From (20), subtracting the classical sum $8rY$ yields

$$S^{(1)} = 4A(r + r^{-1}) - \pi, \qquad \boxed{\vartheta_N^{(1)} = \pi - 8A\chi_2(r) + 4A(r - r^{-1}).} \tag{28}$$

For exchanged arguments the antisymmetric coefficient changes sign. At $r = 1$ the finite-Y antisymmetric phase is zero, whereas the ordered large-Y limit of (28) is $\pi/2$ as $r \uparrow 1$. The near-equal-amplitude boundary layer is therefore not covered by this ordered expansion. The $r \downarrow 0$ limit is also nonuniform: the forward coefficient tends to zero while its symmetric and antisymmetric parts have opposite $1/r$ terms. They must be combined before taking a soft-soliton limit.

(c) The one-body calculation supplies a nontrivial consistency check:

$$C = s_1 - A/2, \qquad s_1 = 3/(4\pi), \qquad C = 1/(2\pi), \qquad A = 2(s_1 - C). \tag{29}$$

Consequently the coefficient of $r + r^{-1}$ in the symmetric phase is $8(s_1 - C)$, agreeing with the independently inverted Jacobian. A common sound-line shift changes C and s_1 equally and leaves this coefficient unchanged. The decomposition found here is an edge dilogarithm, a rational Jacobian term, and an endpoint constant. No additional independent sound-speed two-body term is found in this strict corner. Splitting the rational term into separate multiples of C and s_1 is algebraically possible but is not an independently established separation of mechanisms.

Checks: `twobody_toda_identities.py`.
Checks: `twobody_toda_edge.py`.
Checks: `twobody_toda_analysis.py`.

5 Head-on channel and why reference data must accompany the answer

(c) For opposite edges, $f_2^+ = O(h)$ while $f_2^- \to B(v)/c - 1$. The opposite-edge directional interaction and its edge reference are $N_{12}/h = O(h)$ and $U_{12}/h = O(h)$ at fixed y_j. The leading classical terms are $\delta_T/h \to 4\sqrt{y_1 y_2}$ and $C_{12}/h \to -4\sqrt{y_1 y_2}$. Thus the leading directional residual, its symmetric part and its antisymmetric part all tend to zero. New finite-sea extrapolations test both argument orders. This assertion concerns the stated iterated corner, not finite-amplitude head-on scattering or a global crossing continuation of (28).

(c) The algebraically antisymmetric endpoint representative in the audited frame dictionary is $\mathcal{B}_{12} = \widehat{\Phi}_{12} - \mathfrak{M}_1\mathfrak{Q}_2$. It is unique once this particular subtraction is specified. Its edge values are, however,

$$\mathcal{B}_{\rm ov}/h = H - B_1 B_2 + c(B_1 - B_2) + \pi, \tag{30}$$

$$\mathcal{B}_{\rm ho}/h = (B_1 - c)(B_2 - c). \tag{31}$$

Relative to their classical phases, these contain respectively

$$\mathcal{B}_{\rm ov}/h - \delta_{\rm cl,ov} = 2\sqrt{\pi}\sqrt{Y}(1 - r) + \vartheta_N^{(1)} + O(Y^{-1/2}), \tag{32}$$

$$\mathcal{B}_{\rm ho}/h - 4rY = -2\sqrt{\pi}\sqrt{Y}(1 + r) + \pi + \frac{2}{\pi}(r + r^{-1}) + O(Y^{-1/2}). \tag{33}$$

At fixed small β, $\sqrt{Y} = \beta_1/\sqrt{\hbar_B}$. Discarding these terms silently would change the quantum expansion and, in the head-on channel, the reported constant. The JSON retains U_{12}, U_{21}, the four-term phase and $\mathcal{B}_{12}/h$ as well as R.

(d) Removing an arbitrary antisymmetric addition from the Toda Fredholm problem does not by itself identify any of these representatives with the continuum canonical scattering operator. The frame audit explicitly leaves that endpoint-operator identification open, and the Phase 2i audit does not select an absolute phase. No new choice of regulator, physical frame, statistics offset or crossing branch is made here. Accordingly the table for ϑ_N must not be relabelled as the continuum Gamma-relative ϑ without this additional map.
Checks: `twobody_toda_identities.py`.
Checks: `twobody_toda_scan.py`.
Checks: `twobody_toda_analysis.py`.

6 Numerical extraction and error budget

(c) The finite-sea mesh uses eleven values

$$h \in \{.08, .06, .04, .03, .02, .015, .01, .0075, .005, .00375, .0025\},$$

with $N = \lceil 10/h + 200 \rceil$, plus 25% refinements at $.01, .005, .0025$. The maximum mesh has 5250 points. Its Gauss weights are recomputed from long-double Newton iteration and a Legendre recurrence after the initial SciPy-weight run failed reciprocity; the sea linear algebra remains float64. Polynomial quadrature moments are checked independently. Fifteen pairs $Y = 4, 8, 16, 32, 64$, $r = .4, .5, .6$ are solved in both channels and both argument orders, with exact endpoint q and individually inverted dressed momenta. Every solve checks counting, momentum inversion, endpoint identities and reciprocity. The h fits are evaluated at 60 digits; their float64 inputs are not represented as high-precision sea data.

(c) The central extrapolation is quartic on the last seven h points. Alternatives are cubic on six, quintic on eight, quadratic on four, and cubic plus $h \log h$ or $h^2 \log h$ on seven points. No target intercept is inserted. The h uncertainty is the largest change of intercept. The mesh error is the largest refined-minus-base difference plus $2\,10^{-10}$, propagated through the largest extrapolation

weight norm. The two errors are added. The same method separately checks the vacuum sound coefficient against $3/(4\pi)$.

(c) The half-line grid has 36 pairs: nine scales from 4 to 1024 at $r = .4, .5, .6$, and scales $32, 128, 512$ at $r = .25, .75, .9$. Each convolution uses 32-digit Talbot inversion and geometrically segmented 16- and 24-point mpmath Gauss rules, resolving the $u = O(1)$ layer. De Hoog inversion at 45 digits independently checks four one-body points. The largest cross-method difference is $2.57\,10^{-30}$ and the largest 16/24-node convolution difference is $1.01\,10^{-15}$. The derivative entering the convolution is separately checked by centered differences of the inverted B. Quadrature differences are propagated through the amplitude-fit weights. These diagnostics do not bound a common systematic error of the half-line model.

(c) For the three dense ratio grids, the central large-Y fit is cubic in $1/Y$ through the last four points. Alternatives include quadratic fits, a $Y^{-3/2}$ term, a $Y^{-2}\log Y$ term, and a two-point linear fit. For the three sparse ratio grids the quadratic intercept and the last-two linear intercept are compared. All intercepts are free. Their largest spread is the amplitude-tail error; it is added to the propagated quadrature diagnostic. A separate table compares the finite-sea intercepts against the half-line model rather than merging their errors into a fictitiously rigorous bound on the limiting physical observable.

(c) The following numbers are generated by `twobody_toda_analysis.py` from the new caches.

r	Forward intercept	Tail + quadrature	ϑ_N intercept	Tail + quadrature
0.25	−0.00226183803	$1.79\,10^{-7}$	0.4336115256	0.00033308
0.40	−0.00962285926	$7.65\,10^{-9}$	1.2857724554	$3.09\,10^{-6}$
0.50	−0.01951540939	$6.88\,10^{-10}$	1.5305278134	$1.13\,10^{-6}$
0.60	−0.03545557168	$9.61\,10^{-9}$	1.6631322646	$5.30\,10^{-7}$
0.75	−0.07717786602	$3.83\,10^{-7}$	1.7381235642	$2.23\,10^{-6}$
0.90	−0.16035085391	$2.04\,10^{-6}$	1.7009286895	$3.11\,10^{-6}$

(c) For $r = 1/2$, the proposed exact antisymmetric value is $1.53052781327896308\ldots$. The fit did not use this value.

Y, $r = 1/2$	Sea R_{12}	Mesh error	h-fit spread	Difference from WH
4	−0.0248313244	$8.01\,10^{-9}$	$3.11\,10^{-8}$	$2.62\,10^{-11}$
16	−0.0208546047	$9.47\,10^{-9}$	$1.70\,10^{-6}$	$7.09\,10^{-9}$
64	−0.0198545583	$1.53\,10^{-8}$	0.00011925	$4.16\,10^{-6}$

Y, $r = 1/2$	Head-on R_{12}	Total diagnostic	Head-on R_{21}	Total diagnostic
4	$-6.37\,10^{-11}$	$3.75\,10^{-8}$	$1.16\,10^{-10}$	$1.66\,10^{-7}$
16	$-4.55\,10^{-8}$	$2.16\,10^{-6}$	$7.13\,10^{-8}$	$7.71\,10^{-6}$
64	$-1.92\,10^{-5}$	0.00019692	$3.35\,10^{-5}$	0.00062092

(c) Restricted three-parameter candidate fits to all six independent numerical antisymmetric intercepts give:

Basis	Maximum absolute residual
$1, \chi_2(r), r - r^{-1}$	$1.44\,10^{-5}$
$1, r, r^{-1}$	0.01579042
$1, \log r, \log[(1-r)/(1+r)]$	0.08669543
$1, r - r^{-1}, \log r \log[(1-r)/(1+r)]$	0.02143624

(c) The freely fitted coefficients in the first row are $(3.14175645, -1.27339438, 0.63667460)$, compared with $(\pi, -4/\pi, 2/\pi)$. This tests the specified finite bases; it is not an exclusion theorem for all rational or logarithmic functions. **(c)** This unweighted fit is affected by the less precise $r = 0.25$ intercept. The proposed exact formula is tested separately against each point and its own uncertainty, and all six comparisons pass; the unweighted residual is not its error budget.

(c) The independently extrapolated sound coefficient is $0.238732414641 \pm 8.99\,10^{-9}$, compared with $3/(4\pi) = 0.238732414637843\ldots$. **(c)** Recorded times for completed dense-solve and convolution rows are 953 s and 1136 s respectively. These heavy jobs were run concurrently; the default rerun validates their caches. The failed refinement and subsequent identity checks are not included in these cached-row totals.

(d) The quoted errors are deterministic sensitivity envelopes, not statistical confidence intervals and not a proof that no other asymptotic term is present. In particular, a small contour-inversion difference does not bound finite-sea matching, and no sampled estimate controls $r = 0$ or $r = 1$. Full quantum equivalence is not inferred from agreement of this coefficient with a possible future continuum result.

Checks: `twobody_toda_scan.py`.
Checks: `twobody_toda_edge.py`.
Checks: `twobody_toda_analysis.py`.

7 Verification summary, failures and reproducibility

Claim	Status	Check and limitation
Endpoint evaluation, reciprocity and dressed Jacobian	(c)	`twobody_toda_identities.py`; independent 70-digit quadrature and root derivative.
Correctly referenced edge phase and reverse phase	conditional (c)	`twobody_toda_edge.py`, `twobody_toda_scan.py`; two-term/four-term terms retained.
Forward dilogarithmic coefficient	conditional (c)	Independent integral, direct inverse-Laplace convolution, free intercepts.
Symmetric coefficient and one-body consistency	(c) in edge model	Exact reciprocity and same $A = 2(s_1 - C)$, tested numerically.
Antisymmetric directional function and head-on zero	conditional (c)	Both finite-sea orders and channels; explicit normalization in JSON.
Printed frame-audit four-term PDE	refuted (c)	Endpoint reference differentiation; discrepancy $2c[f(u) - f(v)]$.
Physical scalar phase and Gamma-relative residual	(d)	Endpoint operator map, absolute phase and crossing normalization remain open.
Uniform fixed-nonzero-P limit	(d)	No uniform finite-sea/large-Y or near-diagonal estimate supplied.

Failures recorded.

1. The first finite-sea run failed before producing a row because the node-count attribute `N` shadowed a method of the same name. The method was renamed `directional`; no equation, grid or tolerance was changed.

2. The first 5250-node refinement failed the unchanged $2\,10^{-8}$ reciprocity tolerance. The original assertion did not print the residual, so its exact size was not retained; the failed row was not accepted into the cache. Recomputing the high-order Gauss weights using extended-precision Legendre recurrence addresses the quadrature accuracy rather than relaxing the test. The replacement changes the largest weight by $1.40\,10^{-13}$ and passes with maximum reciprocity residual $2.58\,10^{-11}$ and counting error $1.78\,10^{-15}$. The successful replacement is stored in the cache.

3. The printed four-term Krein source in the frame audit fails the exact reference differentiation. The wrong source is not used, and the discrepancy is retained as a positive refutation check. This does not weaken the Fredholm uniqueness condition.

4. Restricted rational-only and logarithm-only candidate fits are tested against the numerical intercepts; their nonzero residuals are reported in the table and JSON. Their rejection is limited to these tested bases.

5. The PDF skill's artifact-marker helper was not installed in this local workspace (`MODULE_NOT_FOUND`); the authorized local LaTeX build is used directly. An exploratory read also requested a nonexistent historical solver filename; the actual solver sources were then located. Neither operation modified an existing file.

6. The first PDF build had a 5.24-point overfull line containing the entire h grid; moving the grid to display math repaired it. Nonbreaking spaces also keep claim labels with their following text at page boundaries.

No mathematical check was weakened to hide a failure. The success markers certify the asserted checks, not the open physical identifications. All four final scripts printed `ALL CHECKS PASSED`.

Reproduction and interface. Run only the new scripts, in this order:

```
python3 phase2/scripts/twobody_toda_identities.py
python3 phase2/scripts/twobody_toda_scan.py
python3 phase2/scripts/twobody_toda_edge.py
python3 phase2/scripts/twobody_toda_analysis.py
```

The two heavy scripts resume their own JSON caches; `--refresh` regenerates them. The coordinator-facing `twobody_toda_grid.json` contains definitions, signed channels, raw finite-h fit variants, the finite-Y half-line grid, the limiting coefficients, candidate residuals and separate numerical error components. The raw sea and edge caches are also preserved. Build the PDF by running `pdflatex` twice from `phase2/`; the numerical table fragment is supplied alongside this source.

Internal sources read; no new external bibliography. The sources used are `AGENTS.md`, `README.md`, the conclusions and relevant Toda sections of `phase2/report.tex`, `frame_dictionary.tex`, `audit_frame_dictionary.tex`, `audit_phase2i.tex`, `onebody_toda_precision.tex`, `audit_onebody_scheme.tex`, the residual-phase sections of `smx_candidates.tex` and `audit_smx.tex`, and the relevant `errata.tex` and `audit_toda_dictionary.tex` corrections. Implementation sources were inspected for conventions and methods; none of their executable modules was imported or run. All numerical results in this note are new task-prefixed outputs.